EART 11300 Environmental Science Practical

Identification of benthic macroinvertebrates



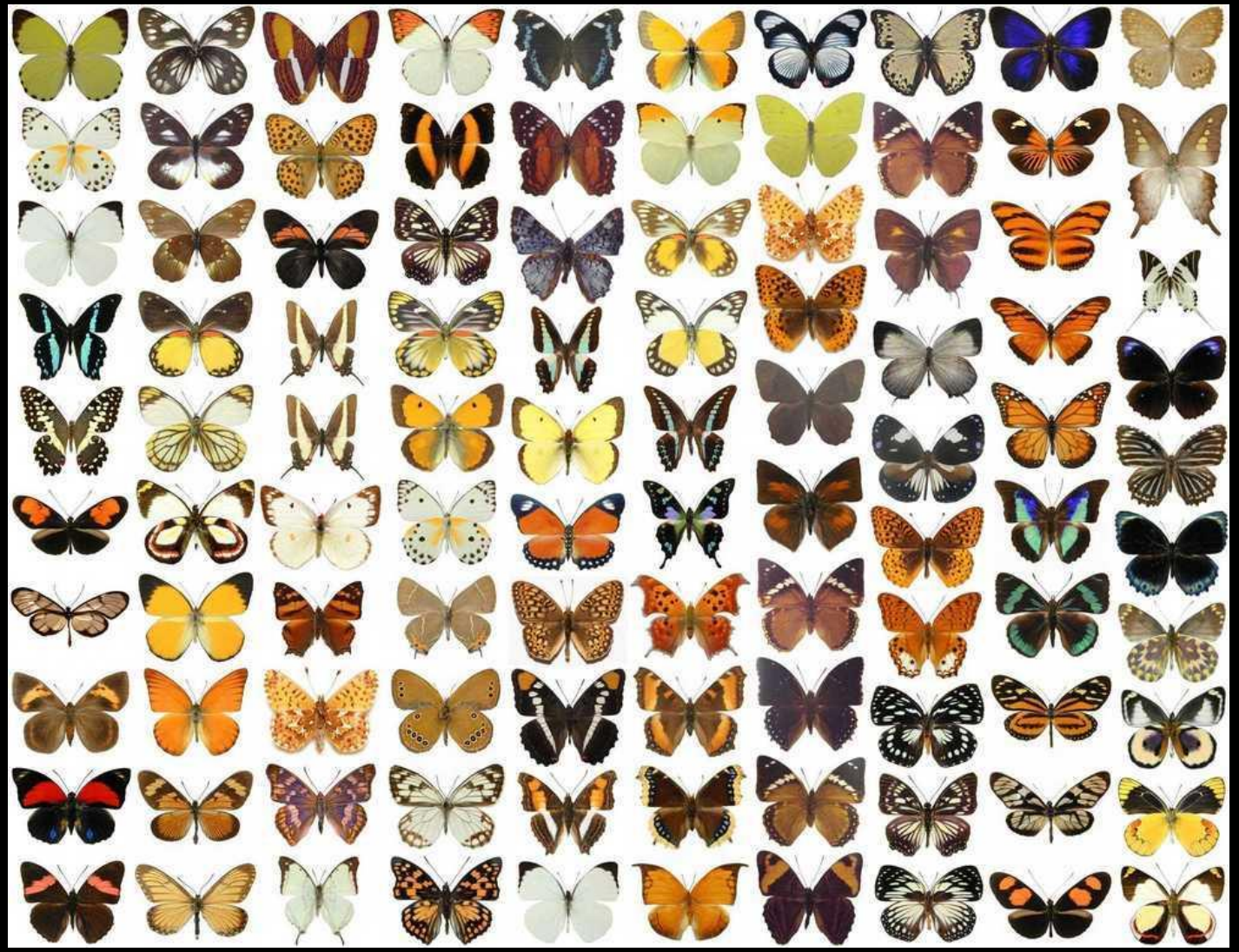
Dr Cecilia Medupin

River Aire at Gordale Beck, Yorkshire Dales

Intended learning outcomes

1. What is taxonomy?
2. What are benthic macroinvertebrates & how are they collected?
3. Identify parts of a dissecting microscope
4. Identify key benthic macroinvertebrates and their characteristics
5. How can river be assessed using benthic macroinvertebrates?

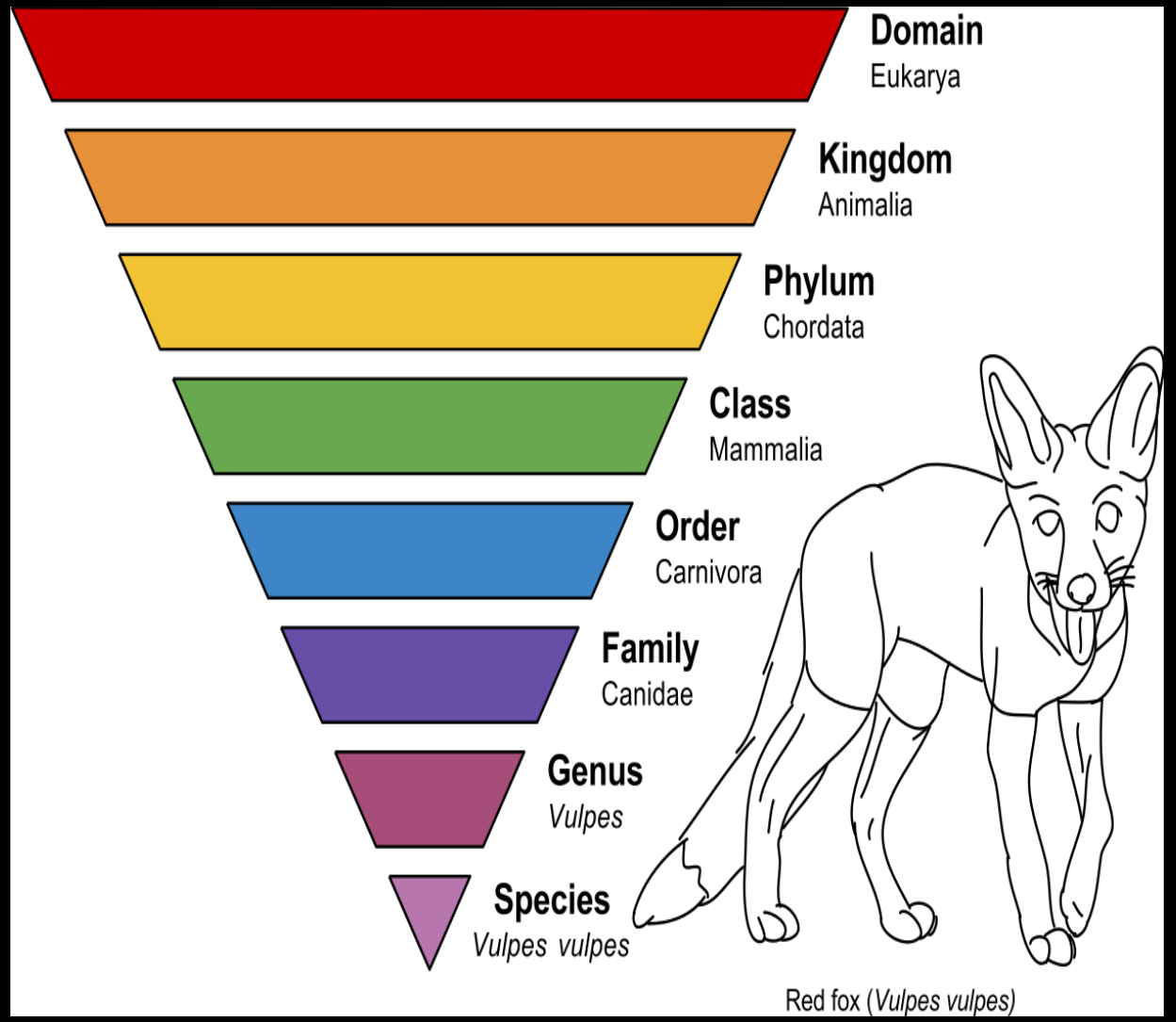
What is taxonomy?



Karl von Linne (1707-1778)

“father” of modern taxonomy





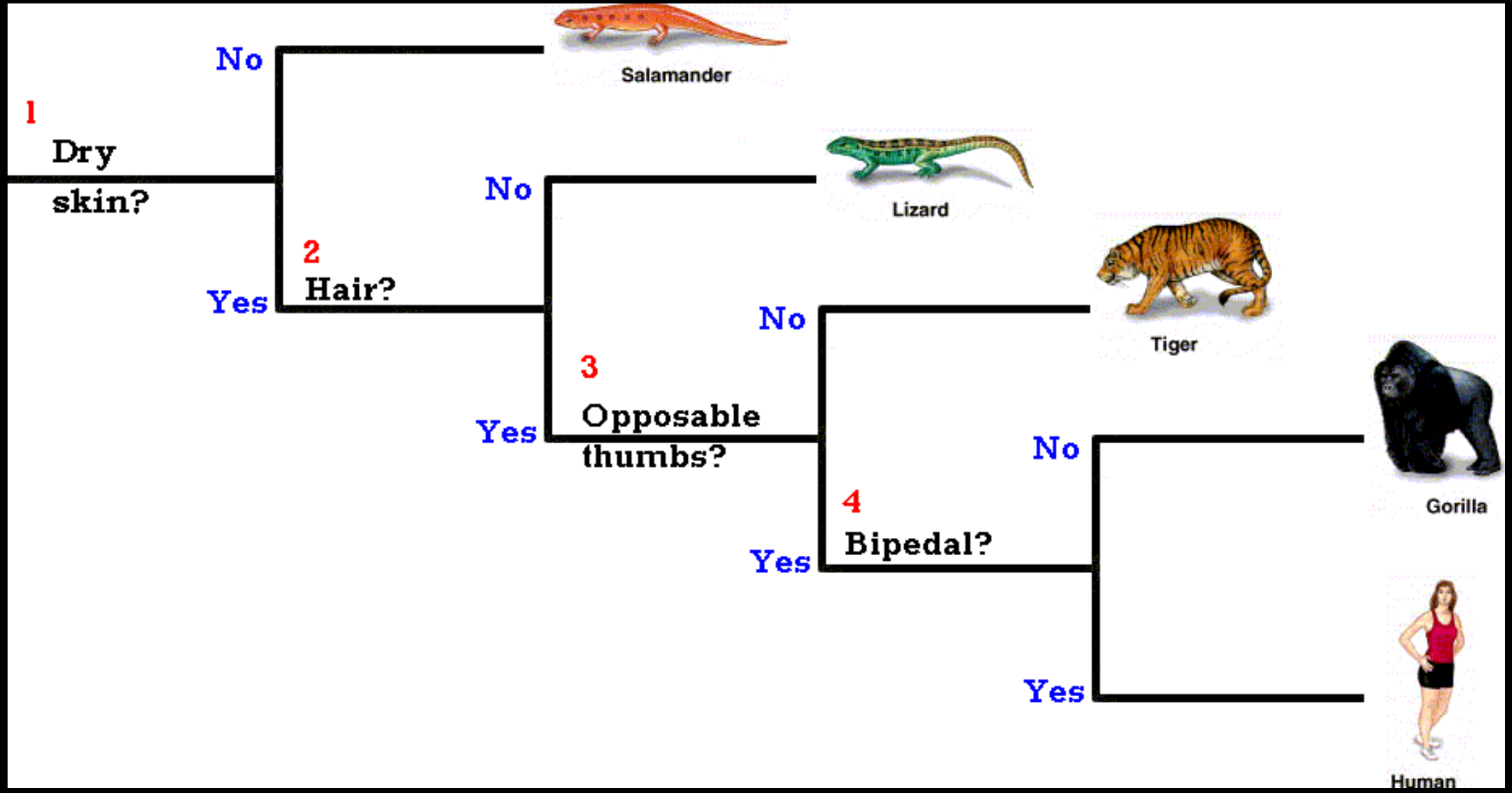
Binomial Nomenclature

The first word is the genus and the second word is the species.

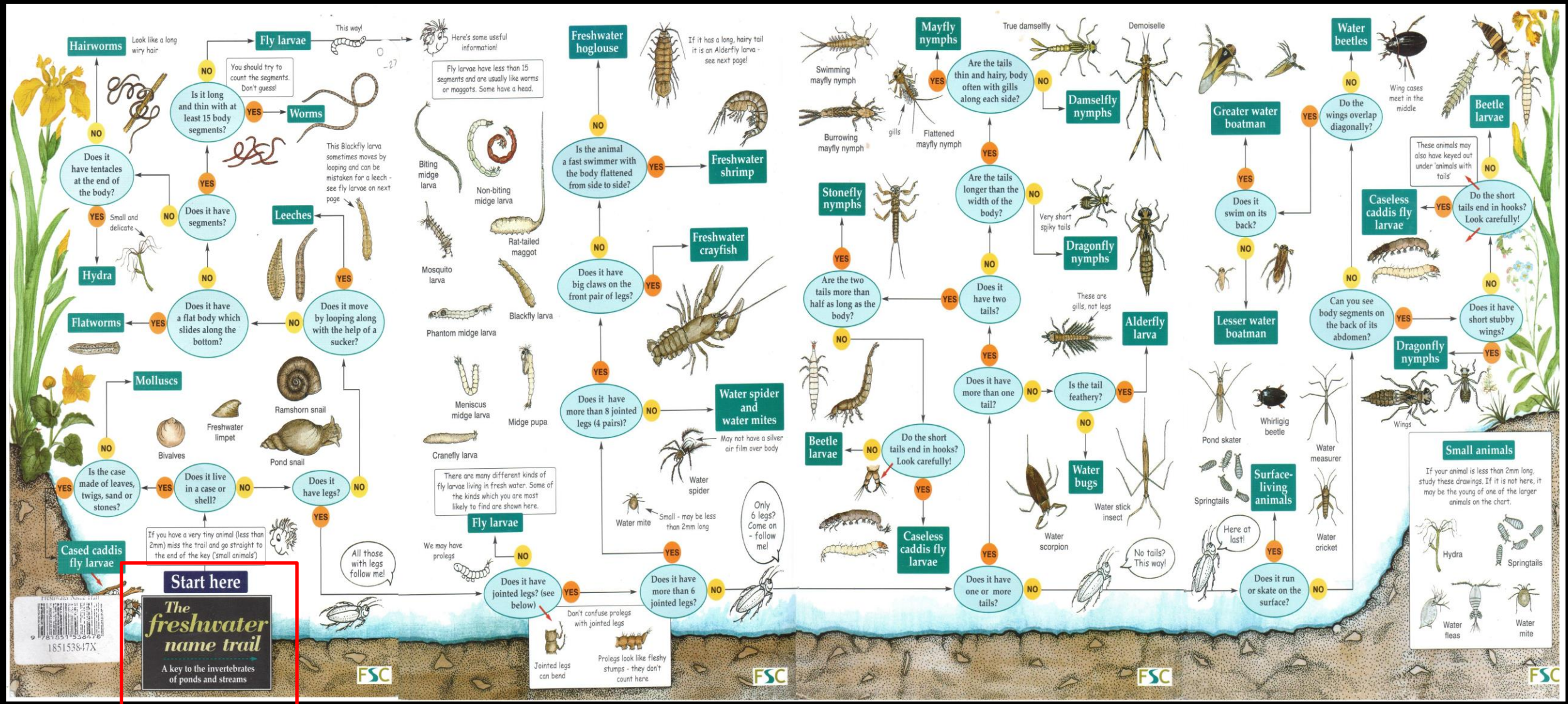
Genus species
Baetis rhodani
Gammarus pulex

Rules:
International Code of Botanical Nomenclature (ICBN), International Commission on Zoological Nomenclature (ICZN)

Dichotomous key



Freshwater Name Trail



Benthic Macroinvertebrates

- Benthic macroinvertebrates- organisms that inhabit bottom substrates for at least part of their life cycle and are retained by a 200µm to 500µm mesh.

- They are key components of aquatic food webs that link organic matter and nutrient resources



Heptageniidae



Gammaridae

Li, Zheng & Liu (2010) Biomonitoring and Bioindicators Used for River Ecosystems: Definitions, Approaches and Trends https://doi.org/10.1016/j.proenv.2010.10.164

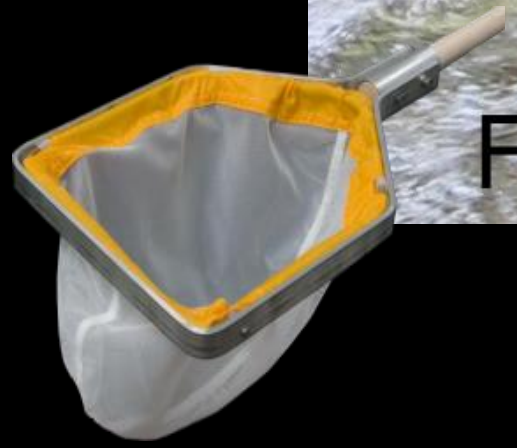
How your samples were collected

- Sampling- riffle, pool, weeds,

Method

- kick sampling using a 1mm mesh –kick net
- 3-minutes' kicknet/sweep
- Empty into buckets half-filled with water
- 1- minute hand search of large stones
- Rub the stones gently into the bucket
- Sort/identification from white trays
- Record samples of invertebrates





Identification
Benthic macroinvertebrates











http://www.visio-tek.net/public/files/16-VISIO-S.pdf

Huge range of benthic macroinvertebrates~ larval forms















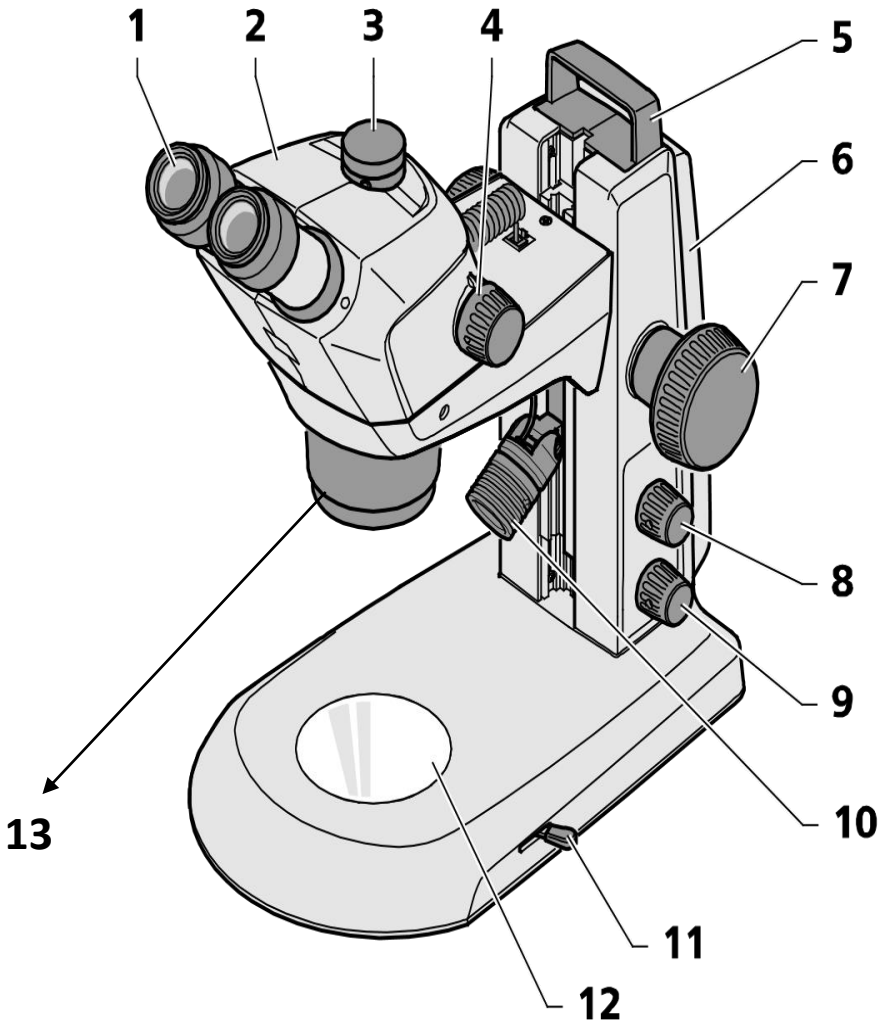




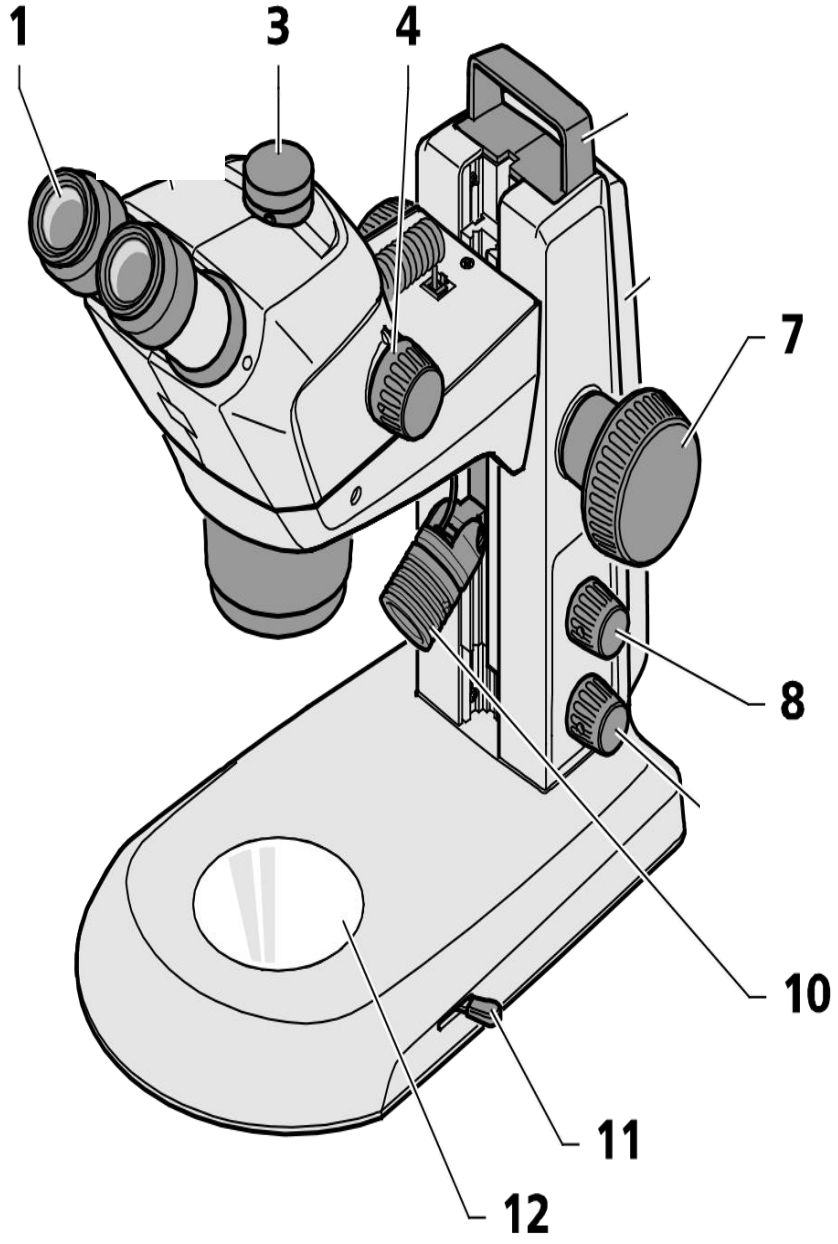




Microscope- Dissecting



1. Eyepiece in eyepiece tube (eyepiece 10x/23)
2. Microscope body with installed vertical illuminator
3. Interface for cameras with connector
4. Zoom button to adjust the magnification
5. Handle
6. Stand
7. Focusing drive (knob) to focus the specimen
8. Push-button for switching ON and OFF the illuminators or the additional reflected-light illumination and for adjusting the illumination intensity
9. Push-button for switching ON and OFF the transmitted-light illumination and for adjusting the illumination intensity
10. Reflected-light illumination
11. Lever for adjusting the transmitted-light illuminator – bright field or dark field on the stand
12. Insert plate for placing the specimen on the stage
13. Lens



Practice Questions

Can you identify the numbered sections of this microscope?

The End, Questions

